

Artificial Intelligence

CSIT — 4th Semester — 2081 — Answer Sheet
Subject Code: CSC266

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. How can you relate synapse, dendrite, and axon in biological neural networks with the elements of artificial neural networks? Create a multi-layer ANN with input layer, hidden layer and output layer. Assume necessary inputs and weights to the ANN and illustrate a single iteration of back propagation algorithm to train the ANN.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. What is Skolem constant? How is Skolemization done during resolution? Represent the following statements into FOPL.[2+3+5] All movies are not hit. Sarangi is a movie. All movies which has good script are hit. Sarangi has a good script but Sarangi is sentimental. There is a movie which is comedy.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. How is informed search different from uninformed search? Create a state space with appropriate heuristics, now illustrate how hill climbing search expands nodes to reach a goal. Modify the state space heuristics and demonstrate when the hill climbing will not be complete

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. What is intelligence? Describe the foundations of AI.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is a rational agent? Justify with example how goal-based agent works?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. How uniform cost search is used to search goal in the state space? Illustrate with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. How can you represent knowledge using scripts? Create a knowledge base using script based on your own assumption.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What is reinforcement learning? Configure an ANN neuron to simulate OR gate.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What is robotics? How machine vision is used in robotics?[3+2]

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Define fuzzy logic? Construct a fuzzy rule base expert system with your own considerations of fuzzy sets.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. How is the minimax algorithm used in game search? Consider state space is defined by a collection of pairs like (A, B) representing paths between states A and B. Construct state space for following and use a minimax algorithm.(A, B), (A, C), (B, D), (D, E), (C, F), (C, G), (D, H), (D, I), (E, J), (F, K), (F, L), (G, M), (G, N).The utilities for states H, I, J K, L, M, N are 1, 3, 2, 6, 3, 4, 1 respectively.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Justify which types of environments resembles following agents.a. Mission Game with fixed 5 states having two playersb. Tesla Driverless Robovan where road conditions are changingc. Game Result Predicting Agent where current prediction state is independent of previous state.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.